

Supplementary Materials for C²GES: Structure-Aware Extractive Summarization of Long Power-System Technical Reports with Typed-Path Graphs

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Contents and Data Files

This supplement separates scientific interpretation from implementation records. It contains four tables referenced by the main manuscript. Machine-readable files are provided under `transferable/`: the non-verbatim sampling frame, aggregate metrics, paired ROUGE-L differences, output-length records, exact sign-flip results, page locators, and the development-only calibration report. Source PDFs and verbatim extracted passages are excluded because third-party redistribution permission has not been established.

Table S1. Rights-Safe Sampling Frame

The complete 40-row sampling frame is supplied as the CSV file `rights_safe_report_metadata.csv` under `transferable/rights_safe_metadata/`.

Quantity	Value
Inspected complete PDFs	40
Declared pages	3200
Retained / excluded reports	27 / 13
Development / test reports	12 / 15
Non-summary candidates	12,924

Table S2. Output-Length Records

The complete 14 condition–budget summary is supplied as `output_length_summary.csv`; per-report values are in `output_length_per_report.csv`. Both files are in the post-run diagnostics subdirectory.

<i>K</i>	Condition	Mean words	Mean chars.	> 100	Table	Max. words
5	Full C ² GES	287.7	1843.6	12	14	140
5	Strict variant	329.0	2099.9	15	16	270
5	Semantic-MMR	184.7	1242.1	5	9	138
5	TextRank	177.0	1125.1	1	7	124
10	Full C ² GES	568.9	3619.5	25	26	270
10	Strict variant	596.9	3822.5	28	26	270
10	Semantic-MMR	354.5	2361.2	7	16	138
10	TextRank	369.0	2329.7	4	18	130

Table S3. Corpus and Implementation Checks

Checked property	Recorded result
Reference/body page-interval overlap	0
Normalized exact candidate/reference matches	0
Normalized common substrings of at least 50 characters	0
Report-condition-budget records / expected records	210 / 210
Selected source references resolved to a valid page	1575 / 1575
Maximum path expansions / configured limit	19,638 / 2,000,000
Maximum qualifying paths / configured limit	16,182 / 250,000
Full/strict score comparisons changed	9774 / 19,008
Report-budget selection sequences changed	28 / 30
Aggregate and paired-contrast recomputation	Exact agreement

Detailed checks are retained with the corresponding code and archived run records. Passing these checks establishes consistency of the reported computation; it does not determine semantic summary quality.

Table S4. Exact Paired Sign-Flip Sensitivity

For each contrast, all $2^{15} = 32,768$ report-level sign assignments were enumerated. The procedure assumes report-level independence and sign exchangeability. Holm adjustment treats the six values as one family.

K	Baseline	Signs +/ - /0	Exact p	Holm-adjusted p
5	Strict variant	7/7/1	0.436768	0.436768
5	Semantic-MMR	14/1/0	0.000305	0.001526
5	TextRank	14/1/0	0.000122	0.000732
10	Strict variant	6/8/1	0.200684	0.401367
10	Semantic-MMR	11/4/0	0.006592	0.026367
10	TextRank	13/2/0	0.009644	0.028931

These post-result values are sensitivity analyses. The main manuscript emphasizes effect sizes, paired distributions, and report-composition intervals.

Exploratory Development-Only Calibration

The complete chronology, configuration space, leave-one-report-out results, and machine-readable records are described in `transferable/POST_UNBLINDING_DEV_CALIBRATION_SUPPLEMENT.md`. A zero path-deletion weight was selected in all 12 development folds; the analysis was not applied to the retained 15-report test set.